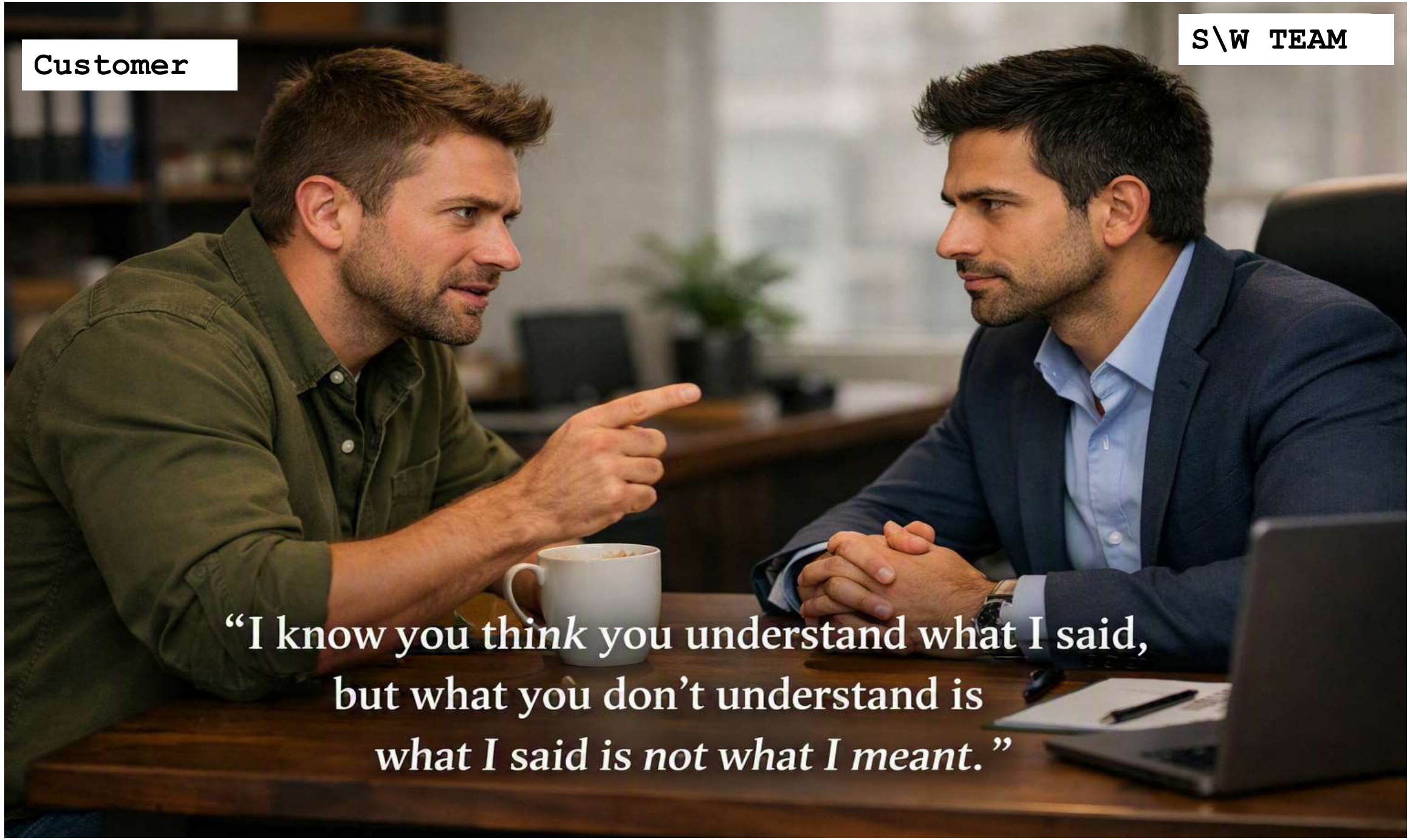

Customer
S\W TEAM
“I know you think you understand what I said,
but what you don’t understand is
what I said is not what I meant.”

What are Requirements?

- A Requirement is:
  - **A capability or condition required from the system.**
- What is involved in requirements analysis and specification?
  - **Determine what is expected by the client from the system. (Gather and Analyze)**
  - **Document those in a form that is clear to the client as well as to the development team members. (Document)**

Understanding and specifying requirements

- **For small problems:** understanding and specifying requirements is rather easy…
- **For industry-standard problems:** Probably the hardest, most problematic and error prone among development tasks…
- The task of requirements specification :
  - **Input**: User needs that are hopefully fully understood by the users.
  - **Output**: Precise statement of what the software will do.

Requirements for Products

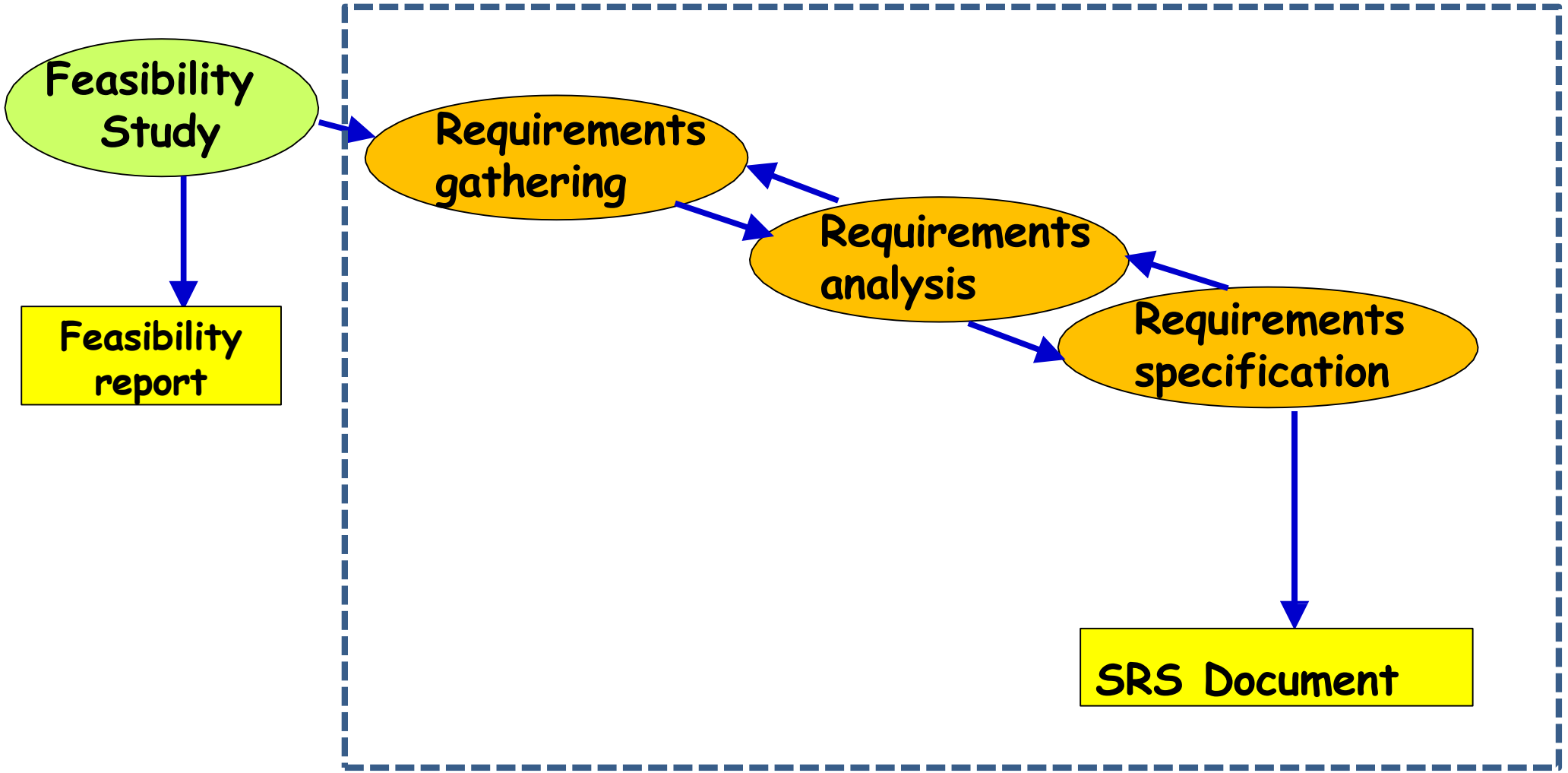
- When a company plans to develop a generic product:
  - Who gives the requirements?
- **The sales personnel!**

Introduction

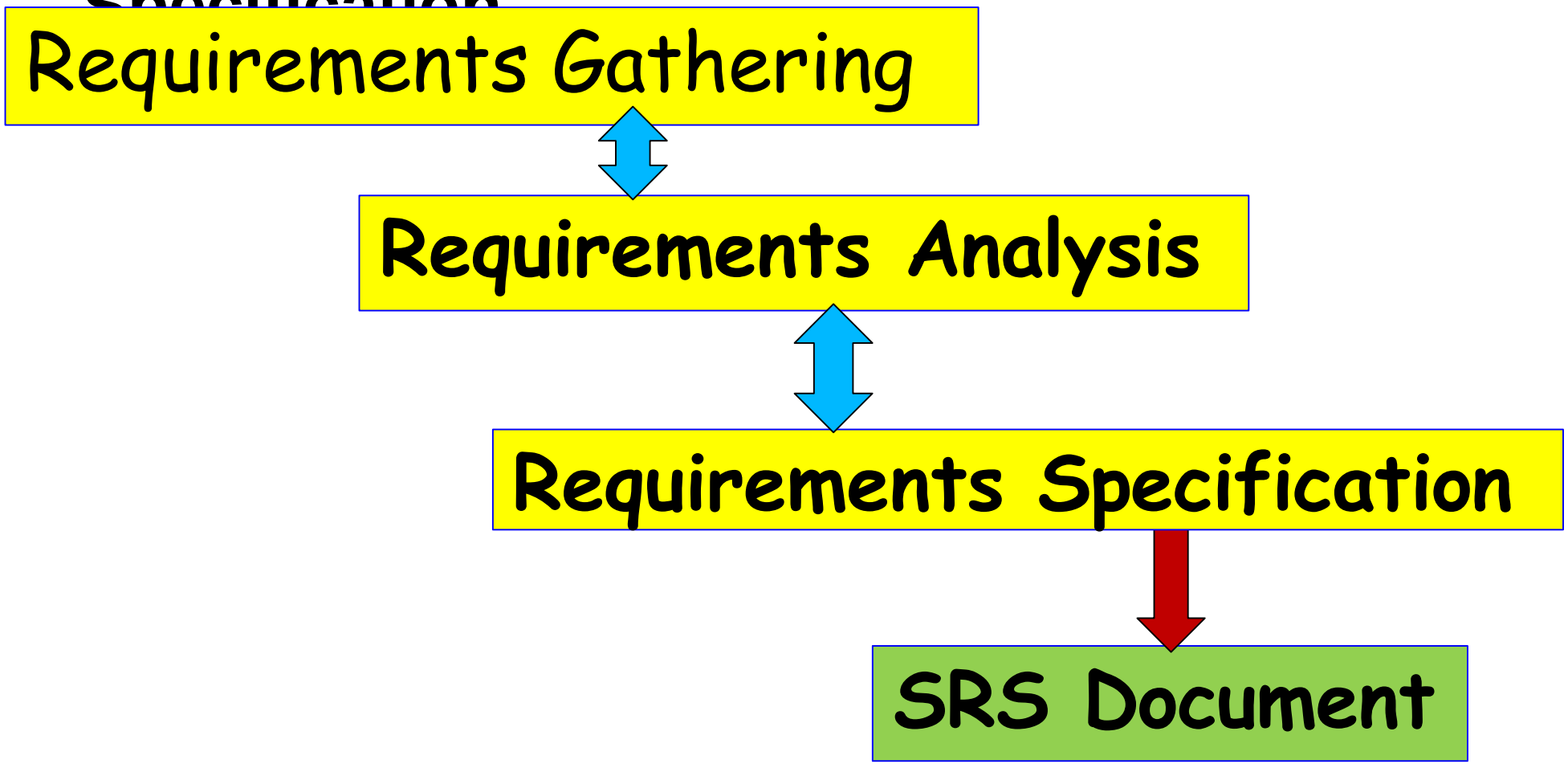
- **The goal of the requirements engineering process is to create and maintain a system(software) requirements document.**

- The overall process includes four high-level requirements engineering sub-processes.
  - **Feasibility Study:** These are concerned with assessing whether the system is useful to the business.
  - **Elicitation and Analysis**: discovering requirements.
  - **Specification:** converting these requirements into some standard form.
  - **Validation:** checking that the requirements actually define the system that the customer wants.

Requirements Engineering Process



Activities in Requirements Analysis and Specification



Requirements Analysis and Specification

- Requirements Gathering:
  - Fully understand the user requirements.
- Requirements Analysis:
  - Remove inconsistencies, anomalies, etc. from requirements.
- Requirements Specification:
  - Document requirements properly in an SRS document.

Requirements Engg.

- Helps s/w engg's to better understand the problem they will work to solve.

- Encompasses the set of tasks that lead to an understanding of
  - What will be the business impact of the s/w,
  - what the customer wants, and
  - how end-users will interact with the s/w.

Who does it?

- *S/w engg's* (sometimes referred to as sys. engg's or analysts in the IT world) and other *project stakeholders* (managers, customers, end-users) all participate in requirements engg.

A Bridge to Design and Construction

- **Designing and building computer s/w is challenging, creative, and just plain fun.**

- **RE, like all other s/w engg. activities, must be adapted to the needs of**

i)the process,
ii)the project,
iii)the product, and
iv)the people doing the work.

- From a software process perspective, RE is a s/w engg. action that begins during the communication activity and continues into the modeling activity.

- It is essential that the s/w team make a real effort to understand the requirements of a problem before the team attempts to solve the problem.

- RE builds a bridge to design and construction.

The journey across the bridge takes us high above the project, allowing

1) s/w team to examine the context of the s/w work to be performed;

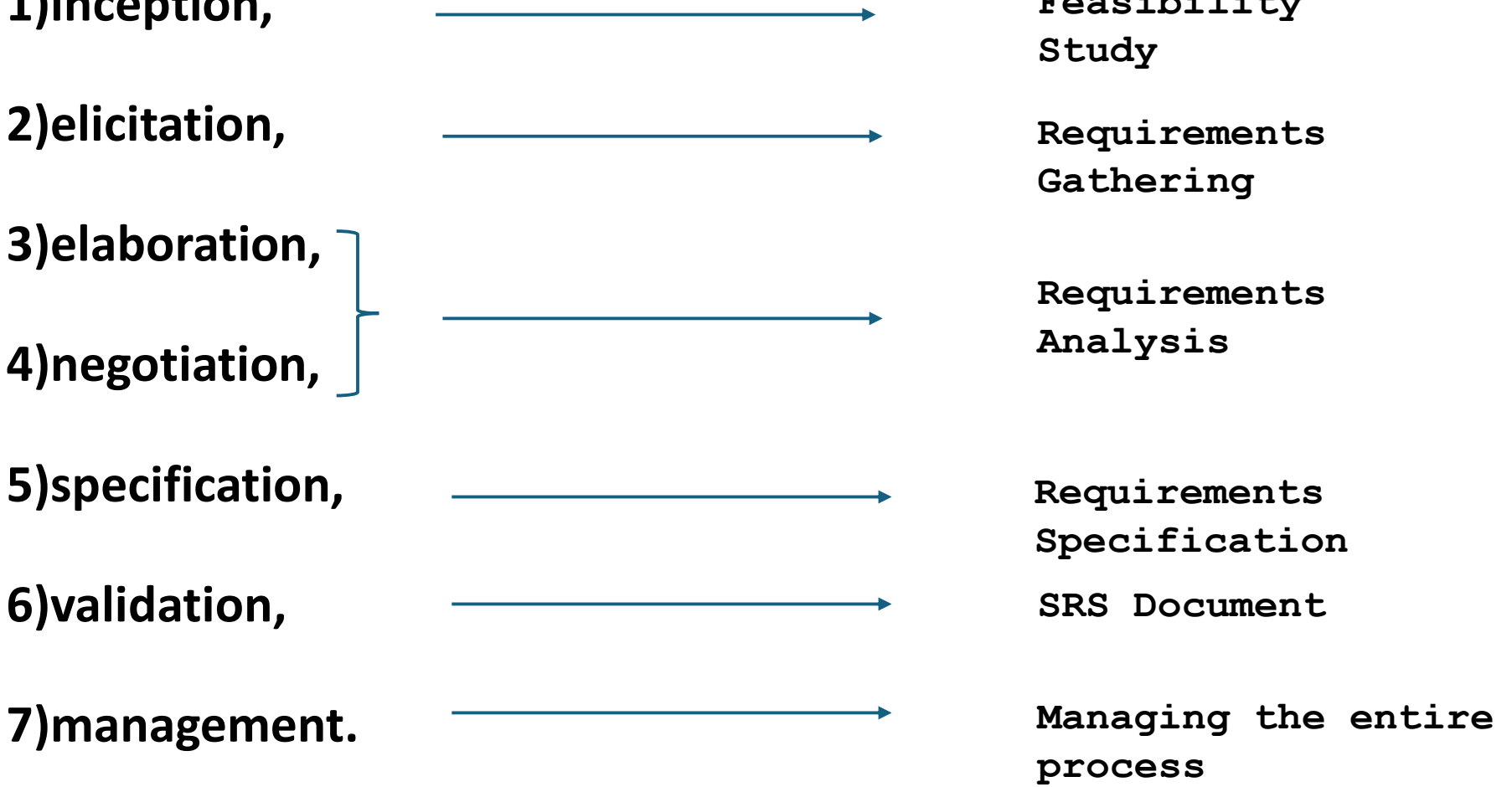
2) the specific needs that design and construction must address;

3) the priorities that guide the order in which work is to be completed; and

4) the information, functions, and behaviors that will have a profound impact on the resultant design.

Requirements Engineering Tasks

The RE process is accomplished through the execution of 7 distinct functions:



➢For all new systems, the requirements engineering process should start with a feasibility study.

➢The input to the feasibility study is a set of preliminary business requirements, an outline description of the system and how the system is intended to support business processes.

➢The results of the feasibility study should be a report that recommends WHETHER OR NOT IT IS WORTH CARRYING on with the requirements engineering and system development process.

1.Inception

- s/w engg's ask a set of context-free questions.
- I. First set(focuses on customer & other stakeholders) of context-free questions:

1)Who is behind the request for this work?

2)Who will use the solution?

3)What will be the economic benefit of a successful solution?

4)Is there another source for the solution that you need?

II.Next set of questions (enables the s/w team to gain a better understanding of the problem):

1)How would you characterize "good" output that would be generated by a successful solution?

2)What problem(s) will this solution address?

3)Can you show me (or describe) the business environment in which the solution will be used?

4)Will special performance issues or constraints affect the way the solution is approached?

III.Final set of questions(focuses on the effectiveness of the communication activity itself):

1)Are you the right person to answer these questions? Are your answers "official"?

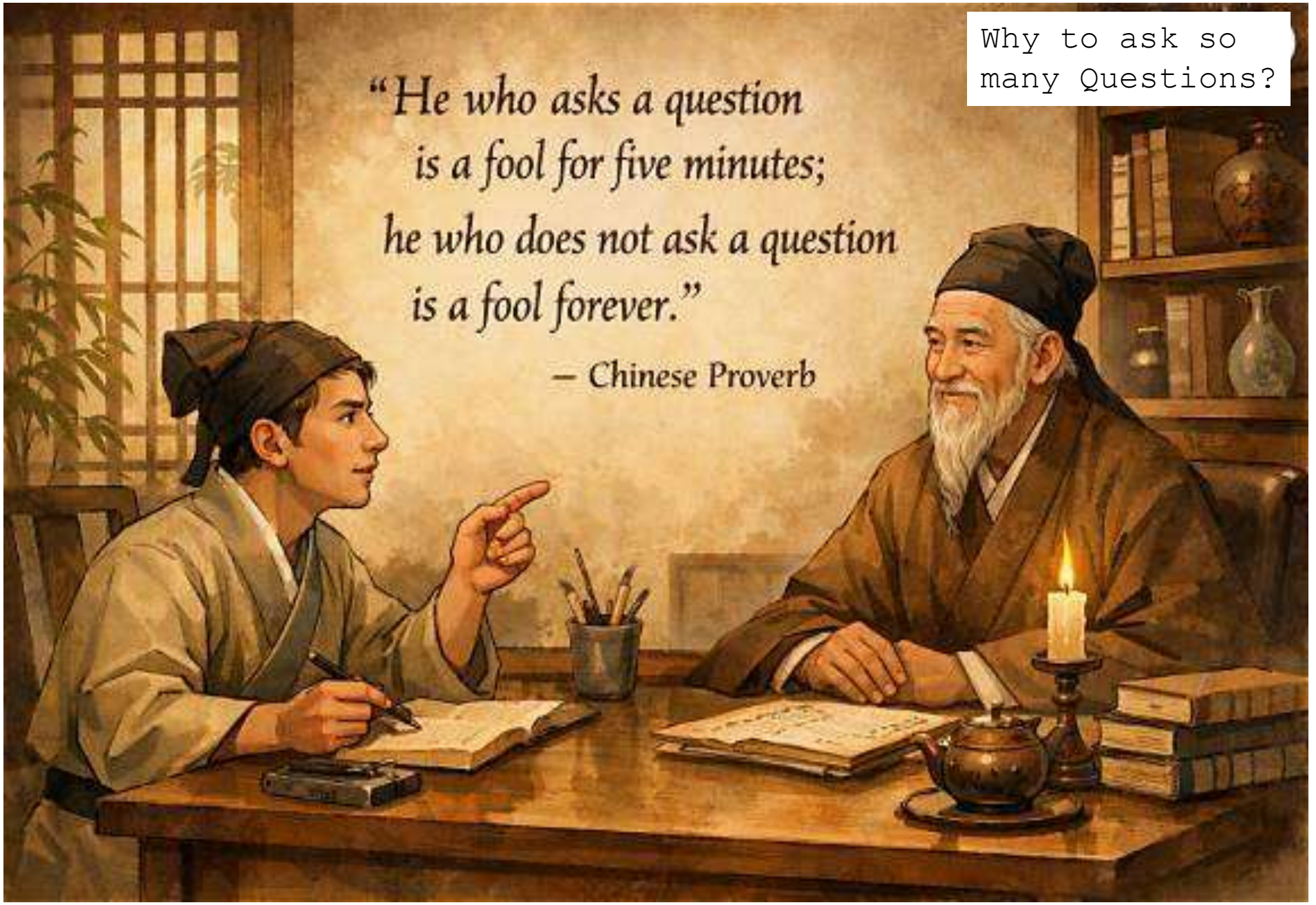
2)Are my questions relevant to the problem that you have?

3)Am I asking too many questions?

4)Can anyone else provide additional information?

5)Should I be asking you anything else?

Why to ask so many Questions?



Intent:

-To establish a basic understanding of the problem,
-The people who want a solution,
-The nature of the solution that is desired, and
-The effectiveness of preliminary communication and collaboration between the customer and the developer.

2.Elicitation

It certainly seems simple enough

—ask the customer, the users, and others what the objectives for the system or product are,

-what is to be accomplished,

-how the system or product is to be used on a day-to-day basis.

- But it isn't simple—it's very hard.

-Christel and Kang identified a no. of problems that help us understand why requirements elicitation is difficult:

i)Problems of scope:

•The boundary of the system is ill-defined or the customers/users specify unnecessary technical detail that may confuse.

ii)Problems of understanding:

•The customers/users are not completely sure of what is needed, have a poor understanding of the capabilities.

•don't have a full understanding of the problem domain,

iii)Problems of volatility:

- The requirements change over time.

- To overcome these problems, RE must approach the requirements gathering activity in an organized manner.

3.Elaboration

- The information obtained from the customer during inception and elicitation is expanded and refined during “elaboration”.

- This requirements engg. activity focuses on developing a refined technical model of s/w functions, features and constraints.

negotiating



4.Negotiation

- It is relatively common for different customers or users to propose conflicting requirements, arguing that their version is "essential for our special needs."

- The requirements engg. must reconcile these conflicts through a process of negotiation.

How Negotiation Works

- **Rough estimation of development effort are made and used to assess the impact of each requirement on project cost and delivery time.**

- **Using an iterative approach, requirements are eliminated, combined, and/or modified so that each party achieves some measure of satisfaction.**

Requirements Elicitation and Analysis

In Requirements Elicitation and Analysis, the software engineers work with customers and system end-users to find out:

- About the application domain.
- What services the system should provide.
- The required performance of the system.
- Hardware constraints

5.Specification:

In the context of s/w, the term specification means different things to different people.

- **It can be**
  1. **a written doc.,**
  2. **a set of graphical models,**
  3. **a formal mathematical model,**
  4. **a collection of user scenarios,**
  5. **a prototype, or any combination of these.**
- **"standard template" should be developed and used for a specification.**
- **It describes the function and performance of a computer-based system and the constraints that will govern its development.**

6.Validation

- The work products produced as a consequence of RE are assessed for quality during a validation step.

- It examines the specification to ensure that all s/w requirements have been stated unambiguously, that inconsistencies, omissions, and errors have been detected and corrected

Why Requirement validation matters

- Requirements validation is concerned with showing that the requirements actually define the system that the customer wants.
- Requirements validation is important because errors in a requirements document can lead to extensive rework costs when they are discovered during development or after the system is in service.
- The cost of fixing a requirements problem by making a system change is much greater than repairing design or coding errors.

The requirements validation process performs various checks include:

- Validity checks
- Consistency checks
- Completeness checks
- Realism checks
- Verifiability checks

Requirements validation techniques

- **Requirements reviews**: The requirements are analyzed systematically by a team of reviewers.
- A requirements review is a manual process that involves people from both client and contractor organizations.
- They check the requirements document for anomalies and omissions.
- Requirements reviews can be informal or formal.

- **Prototyping:** In this approach to validation, an executable model of the system is demonstrated to end-users and customers. They can experiment with this model to see if it meets their real needs.

- **Test-case generation**: Requirements should be testable. If the tests for the requirements are devised as part of the validation process, this often reveals requirements problems.

7.Requirements management

- It is a set of activities that help the project team identify, control, and track requirements & changes to requirements at any time as the project proceeds.

- Requirement Management begins with identification.

- Each requirement is assigned a unique identifier. (ex:REQ-01,REQ-02…)

- Once requirements are identified, traceability table(TT) are developed, each TT relates requirements to one or more aspects of the sys. or its environment.

Requirement traceability

Traceability is the property of a requirements specification that reflects the ease of finding related requirements.

There are three types of traceability information that may be maintained:

- Source traceability information links the requirements to the stakeholders who proposed the requirements.
- Requirements traceability information links dependent requirements within the requirements document.
- Design traceability information links the requirements to the design modules where these requirements are implemented

Requirements traceability

Req id	1.1	1.2	1.3	2.1	2.2	2.3	3.1	3.2
1.1		D	R					
1.2			D			R		D
1.3	R			R				
2.1			R		D			D
2.2								D
2.3		R		D				
3.1								R
3.2							R	

Requirement change managements

- Requirements change management should be applied to all proposed changes to the requirements.

- The formal process for change management brings a controlled way for changing requirements.

- There are three principal stages to a change management process:
  - Problem analysis and change specification:
  - Change analysis and costing
  - Change implementation

Value of a Good SRS

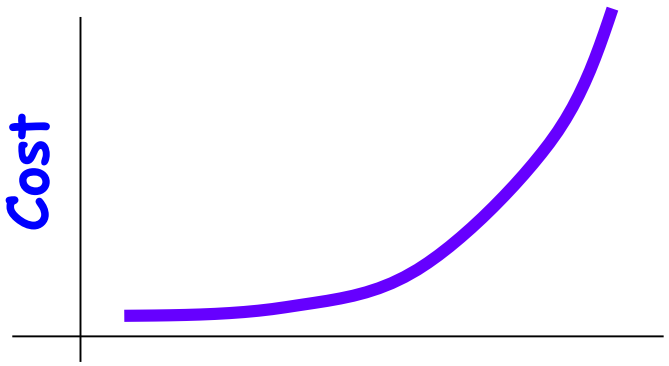
- **A high-quality SRS is a prerequisite to high-quality software.**
- **A high-quality SRS reduces the development cost.**

What are the Uses of an SRS Document?

- Establishes the basis for agreement between the customers and the suppliers
- Forms the starting point for development.
- Provide a basis for estimating costs and schedules.
- Provide a basis for validation and verification.
- Provide a basis for user manual preparation.
- Serves as a basis for later enhancements.

Need for SRS…

- **Good SRS reduces development cost:**
  - Req. errors are expensive to fix later
  - Req. changes cost a lot (typically 40% of requirements change later)
  - Good SRS can minimize changes and errors
  - **Substantial savings --- effort spent during req. saves multiple times that effort**
- **An Example:**
  - Cost of fixing errors in req. , design , coding , acceptance testing and operation increases exponentially



Forms A Basis for User Manual

- The SRS serves as the basis for writing User Manual for the software:
  - **User Manual: Describes the functionality from the perspective of a user --- An important document for users.**
  - Typically also describes how to carry out the required tasks with examples.

SRS Document: Stakeholders

- SRS intended for a diverse audience:
  - **Customers and users use it for validation, contract, ...**
  - **Systems (requirements) analysts**
  - **Developers, programmers to implement the system**
  - **Testers use it to check whether requirements have been met**
  - **Project Managers to measure and control the project**
- Different levels of detail and formality is needed for each audience
- Different templates for requirements specifications used by companies:
  - Often variations of **IEEE 830**

- Requirement Process consists of 3 basic tasks:
  - **1. Problem or Requirement Analysis**
  - **2. Requirements Specification**
  - **3. Requirements Validation**

Problem Analysis

- Basic purpose of this activity is to obtain a thorough understanding of what the software needs to provide.
- Analyst will have a series of meetings with the clients and end users
- In the early meetings, the clients and end users will explain to the analyst about their work, their environment, and their needs as they perceive them.
- In these early meetings, the analyst is basically the listener, absorbing the information provided.

- Once the analyst understands the system to some extent, he uses the next few meetings to seek clarifications of the parts he does not understand.

- In the final few meetings, the analyst essentially explains to the client what he understands the system should do and uses the meetings as a means of verifying if what he proposes the system should do is indeed consistent with the objectives of the clients.

Requirements Specification

- The understanding obtained by problem analysis forms the basis of requirements specification, in which the focus is on clearly specifying the requirements in a document.
- The final output is the SRS document.
- The SRS is written based on the knowledge acquired during analysis.
- Desirable Characteristics of an SRS
  - 1.Correct
  - 2. Complete
  - 3. Unambiguous
  - 4. Verifiable
  - 5. Consistent
  - 6. Ranked for importance and/or stability

Requirements Validation

- Requirements validation focuses on ensuring that what have been specified in the SRS are indeed all the requirements of the software and making sure that the SRS is of good quality. The requirements process terminates with the production of the validated SRS.

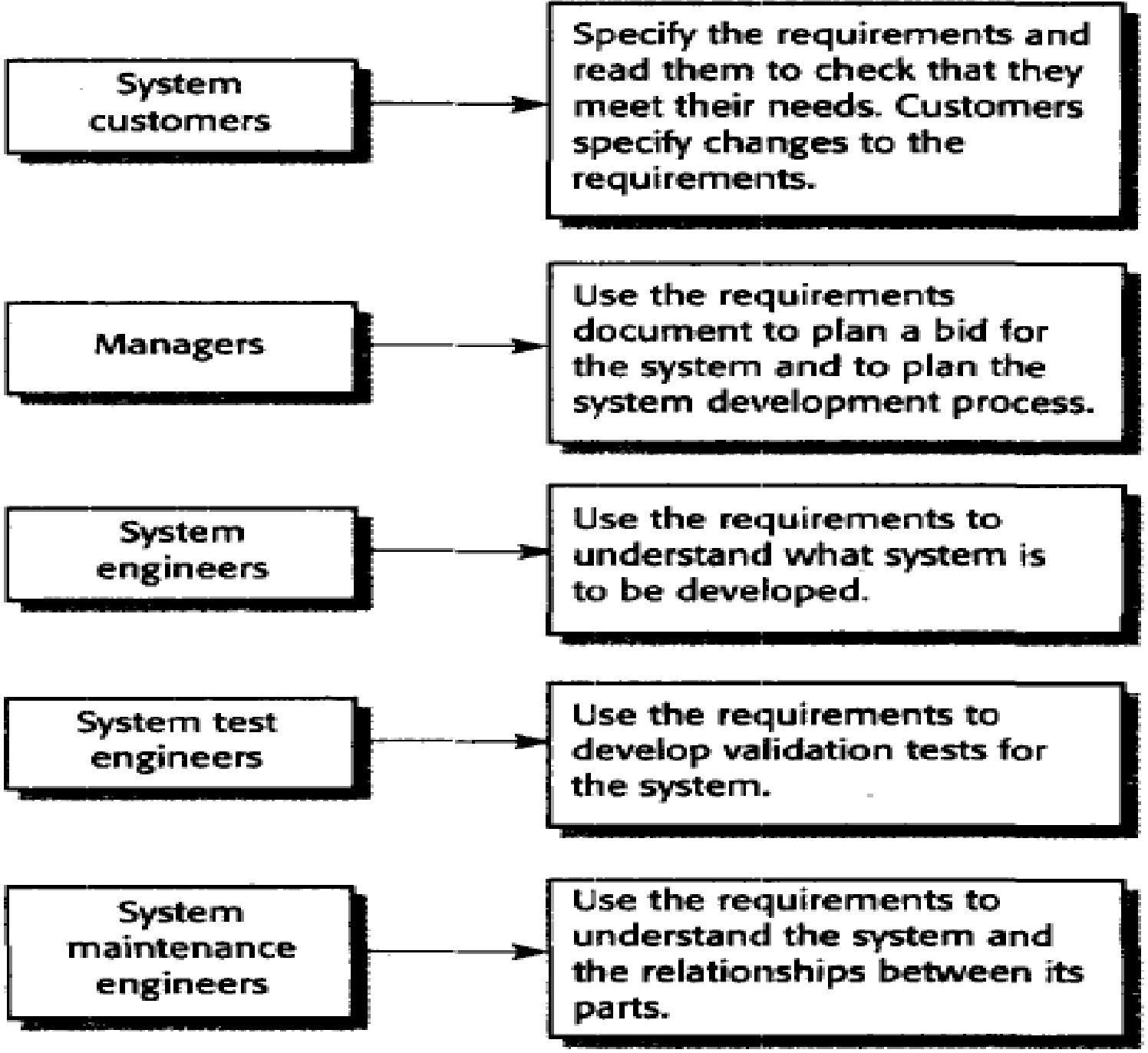
Components of an SRS

- – Functionality
- – Performance
- – Design constraints imposed on an implementation
- – External interfaces

The Software Requirements Document

- The software requirements document or software requirements specification (SRS) is the official statement of what the system developers should implement.

- It should include both the user requirements for a system and a detailed specification of the system requirements


System customers
Specify the requirements and read them to check that they meet their needs. Customers specify changes to the requirements.
Managers
Use the requirements document to plan a bid for the system and to plan the system development process.
System engineers
Use the requirements to understand what system is to be developed.
System test engineers
Use the requirements to develop validation tests for the system.
System maintenance engineers
Use the requirements to understand the system and the relationships between its parts.

Structure of a Requirements Document

1. Introduction
   1.1 Purpose of the requirements document
   1.2 Scope of the product
   1.3 Definitions, acronyms and abbreviations
   1.4 References
   1.5 Overview of the remaining document
2. General description
   2.1 Product perspective
   2.2 Product functions
   2.3 User characteristics
   2.4 General constraints
   2.5 Assumptions and dependencies
3. Specific requirements
4. Appendices
5. Index

3. Specific Requirements(Detailed Requirements)
   3.1 External Interface Requirements
      3.1.1 User Interfaces
      3.1.2 Hardware Interfaces
      3.1.3 Software Interfaces
      3.1.4 Communication Interfaces
   3.2 Functional Requirements
   3.3 Performance Requirements
   3.4 Design Constraints
   3.5 Attributes
   3.6 Other Requirements